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Module 4: Software Design
Session 2: UI and UX Design, Wireframes and Prototypes

Software Developer Level 4

What you will be able to do

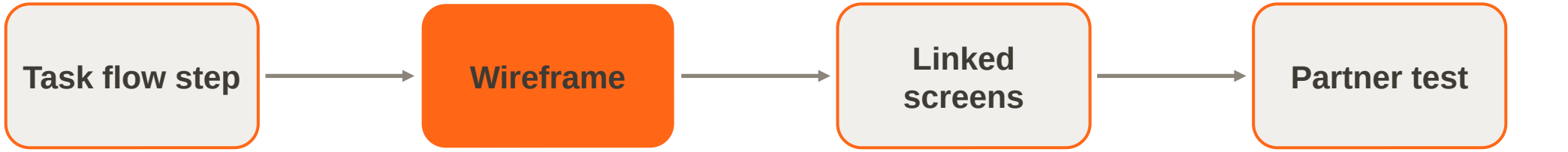
By the end of this session you will be able to:

1. Analyse user needs and accessibility requirements from personas and request data.
2. Design wireframes and a clickable low-fidelity prototype that cover normal, error and emergency paths.
3. Evaluate a design against an accessibility checklist and justify the changes you make.

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From this morning's task flow to screens

This morning you mapped how a tenant reports a repair. This afternoon each step becomes a screen.



What the tenant does

What they see on a phone

What they tap to move on

Where they get stuck

By the end of the afternoon: linked screens for reporting a repair, clicked through by a partner.

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Your Prepare question

CHAT

What made the worse form harder to use?

Type one specific thing in the chat: the form, what you tried to do and what got in your way.

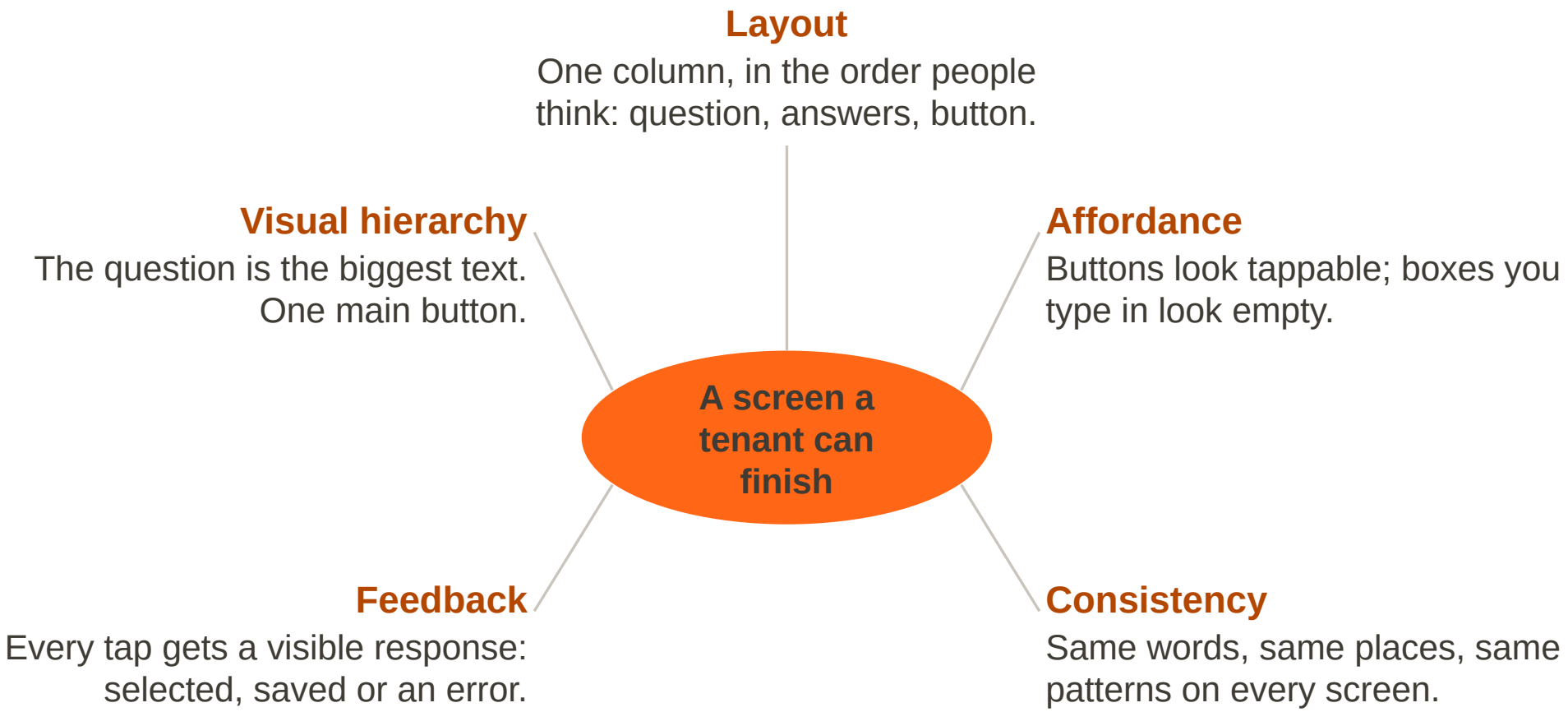
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UI and UX principles

What makes a phone screen usable, shown on a different journey: a tenant changing how Kestrel Moor contacts them.

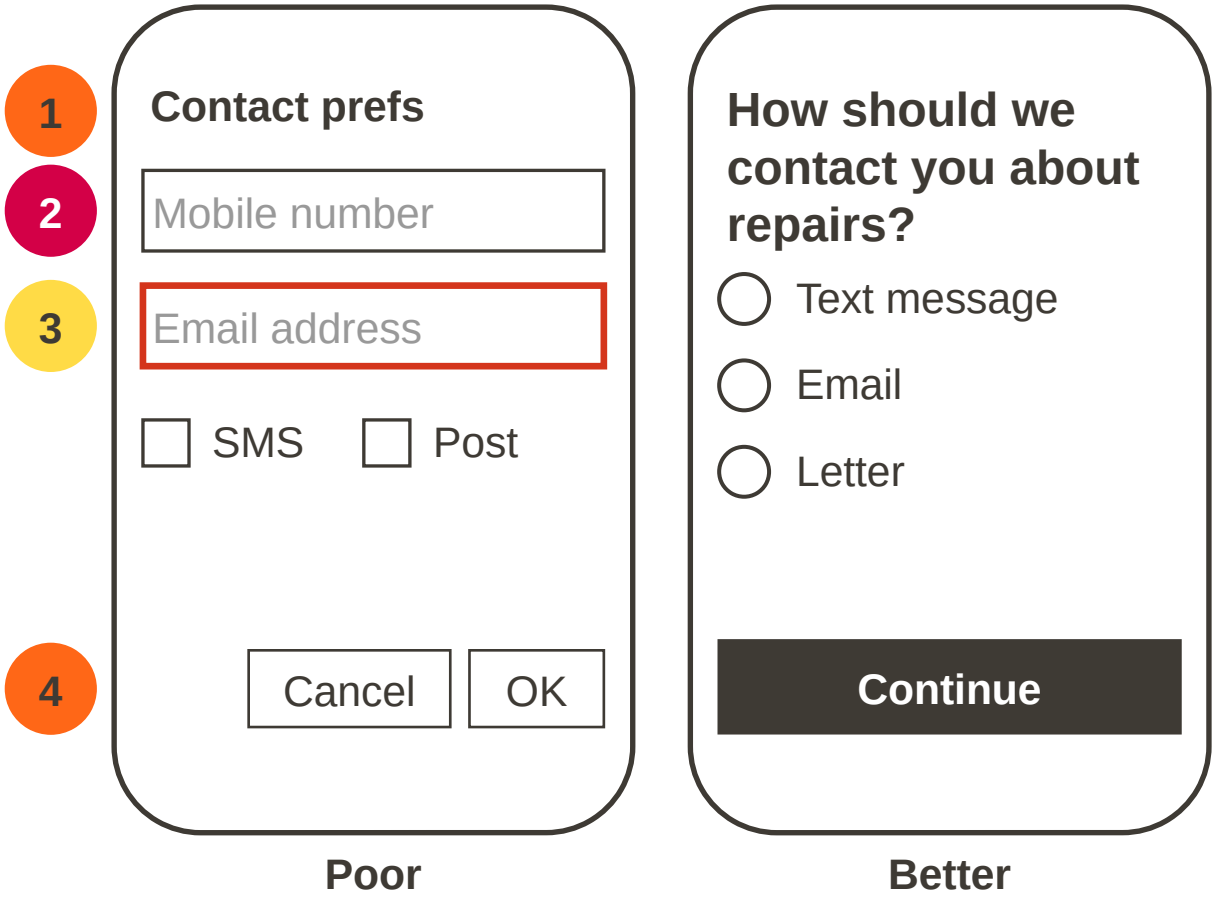
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Five principles behind a usable screen



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Contact preferences: poor and better



1. 'Prefs' and 'SMS' are our words. Ask a question in the tenant's words.
2. Grey hint text vanishes as you type. Put a visible label above each box.
3. A red border alone says nothing to many people. Say what is wrong in text.
4. Small buttons side by side get mis-tapped. Use one full-width main button.

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One question per screen

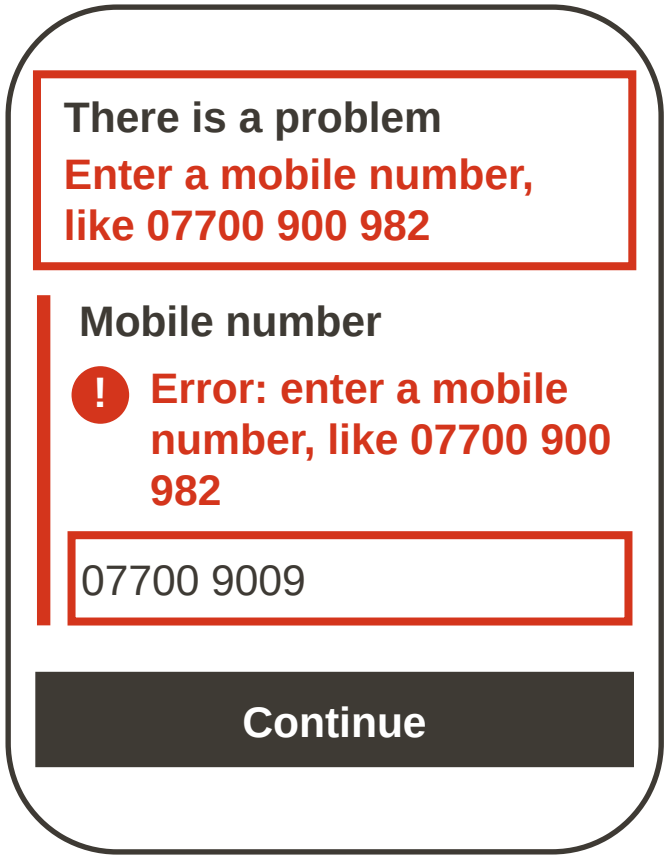
PATTERN

> **On a phone, ask one thing per screen.**

Each answer is easier to give, check and correct, and an error has only one place to point to.

GOV.UK's guide to structuring forms says to start with one thing per page.

An error state, worked on a different screen



1. A summary at the top, headed 'There is a problem', links to each field with an error.
2. The message by the field says what to do, in plain words, with an example.
3. A red bar and border, an icon and the word Error: colour is never the only signal.
4. What the tenant typed is still there, so they correct it rather than start again.

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An emergency interrupts the flow

Where the interruption sits decides whether an emergency waits in a queue.

TOO LATE

Warn after the form is sent

The tenant has typed, added photos and pressed submit.

The report then waits in a queue until someone reads it.

A warning on the confirmation page comes after the risk.

IN TIME

Stop before the form starts

The first screen asks about emergencies, before any sign-in.

An emergency answer leads straight to a stop screen.

It says how to call now, by voice or text relay. Nothing to submit.

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Accessibility and inclusive design

Kestrel Moor's seven-point standard, and designing from evidence about real tenants.

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Kestrel Moor's standard, point by point

Points 1 to 6 map to WCAG 2.2 criteria; point 7 is Kestrel Moor's own rule.

The standard says	WCAG 2.2 success criterion	Level
1. A visible label on every field	3.3.2 Labels or Instructions	A
2. Errors in text, saying how to fix them	3.3.1 Error Identification, 3.3.3 Error Suggestion	A, AA
3. Colour never the only signal	1.4.1 Use of Color	A
4. Text contrast 4.5:1 or more	1.4.3 Contrast (Minimum)	AA
5. Tap targets 24 by 24 CSS pixels	2.5.8 Target Size (Minimum), new in 2.2	AA
6. No re-entry; input kept after an error	3.3.7 Redundant Entry, new in 2.2	A
7. Plain English, one question per screen	Kestrel Moor's own rule	Beyond AA

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Contrast you can measure

4.5:1

Minimum contrast for text in the standard

Maureen Hughes: "Half the forms we get have tiny grey writing."

Light grey #999999 on white: 2.85:1. Fails.

Mid grey #767676 on white: 4.54:1. Just passes.

This dark text on white: 11.29:1.

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Personas are evidence, not decoration

Joan Whitaker, 63

Text size turned up; reading glasses.

Reported mould three times and explained it again each time.

Wants to report once, show what it looks like and know when someone is coming.

Each line is a reason for a design decision. Joan and Amir are two tenants, not all of them.

Photo: an illustrative tenant reporting mould by a kitchen window.



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From evidence to a design rule, worked

Worked on contact preferences. Activity 1 asks for three repair needs.

ONE ROW OF A USER-NEEDS TABLE

(1) User need: hear about my repair in a way I can use, including by letter

(2) Evidence: Maureen Hughes (Case Brief Part 2): "some won't use an app, full stop"; Phase 1 scope: updates by text, email or letter

(3) The design must: offer text, email and letter as equal choices, with none pre-selected

1. Written as what the user must be able to do. A dropdown is a feature, not a need.
2. A source anyone can check: a persona line, an interview or a request_id.
3. Testable: a reviewer can look at a screen and say yes or no.

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Activity 1: Who are we designing for?

Pairs, in breakout rooms

Find the needs the repair screens must meet.

1. Re-read Joan and Amir in the Activities document: what must each of them be able to do?
2. In the data, filter vulnerability_flag to Y and read every vulnerability_note.
3. Fill rows 1 to 3: two accessibility needs, and one need that only the data shows.
4. Write each need as what the tenant must be able to do, and name its evidence.

Extension: see the Activities document

TIME
14 minutes working time

YOU NEED
Activities document: Joan, Amir and the needs table
Your own copy of the Repair Requests data

CORE OUTPUT
Rows 1 to 3: each need, with its evidence (a persona line or a request_id).

Debrief: Which need did the data show that the personas did not?

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Break

20 minutes

After the break you build in draw.io. Download Starter Files.zip from the Files tab and open app.diagrams.net now.

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From wireframe to clickable prototype

Draw the screens, link them, then let someone else click through.

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Why screen 3 matters

2

of 5 gas smells in the sample came in online and were triaged Routine

KMR-26-0019, web form: 28.4 hours to triage.

KMR-26-0112, email: 53.8 hours to triage.

The 3 reported by phone were Emergency, triaged within 0.4 hours.

Case Brief Part 4: anyone who smells gas calls 0800 111 999 first.

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What goes in a low-fidelity wireframe

Low fidelity keeps the conversation on the flow and the words, and keeps changes cheap.

Put in

Real labels and real words

The order of questions and buttons

Error and emergency screens, not only the main path

A short note on each decision and its reason

Leave out

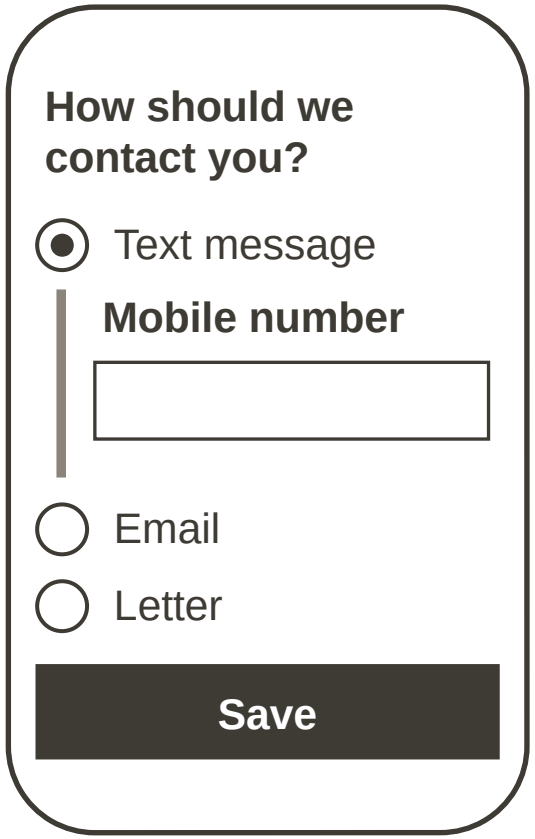
Brand colours, logos and photos

Exact fonts and pixel spacing

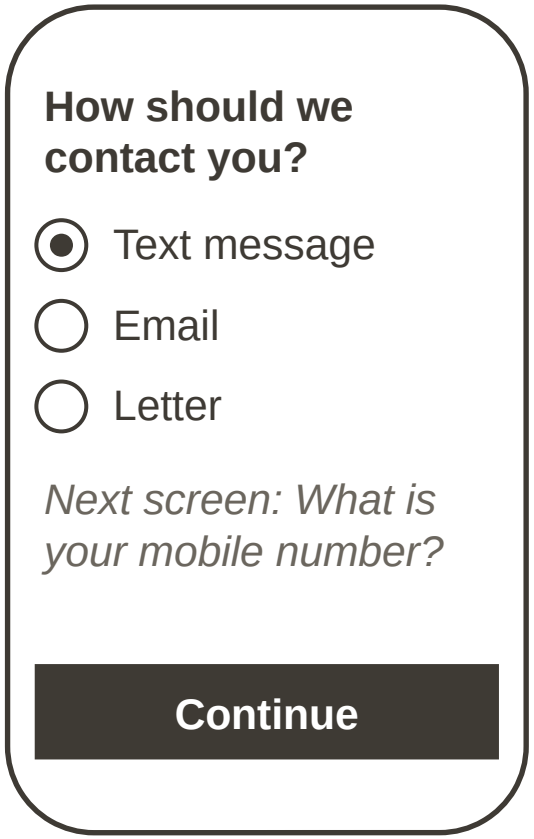
Dummy text: write the real question instead

Decoration that does not help anyone finish

Two layouts before you choose



Option A



Option B

Option A: reveal the field

Choosing Text message opens the number box under it. One screen that grows.

Option B: next screen

Choose, then Continue, then a screen that asks only for the number.

Chosen: A

The number belongs to that choice, so changing one setting takes one step. The reason goes on the wireframe as a note.

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Demo: from blank frame to clickable prototype

1. **Open your own copy**
   app.diagrams.net, then File, Open from, Device. One page per screen.

2. **Copy from Components**
   Paste boxes, labels and buttons into the phone frame. Duplicate a page to reuse it.

3. **Annotate**
   A note shape beside the frame: the decision and the reason.

4. **Link a shape to a page**
   Right-click the button, Edit Link, choose the page (second option).

5. **Test the click-through**
   Hover over the button and click the link: you land on that page.

6. **Save and post**
   File, Save as, Device. Post the .drawio in your breakout chat.

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Activity 2: Prototype the repair journey

Individual, in your pair's breakout room

Design and link the screens for reporting a repair.

1. Screen 1 (no sign-in) asks about emergencies first. Sketch two layouts; choose one, note why.
2. Screen 2 asks one question: what is wrong? Duplicate the page and draw its error state.
3. Screen 3, the stop: who to call, call buttons, Relay UK, safety advice, a way back, no Submit.
4. Screen 4 confirms the report with a reference number.
5. Annotate screens 1 and 3: each note cites a need from your Activity 1 table.
6. Link the screens and click through. Save, then post the .drawio in your breakout chat.

Extension: see the Activities document

TIME

40 minutes working time

YOU NEED

Your own copy of the starter .drawio
Your Activity 1 needs table
Activities document: Case Brief Part 4 extract

CORE OUTPUT

Screens 1 to 4 linked, with screen 2's error state and two annotations.

Debrief: Show your screen 3: what does a tenant who cannot make a voice call do next?

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A review comment someone can act on

Four parts, every time:

1. Which screen
2. Which point of the standard
3. What fails, as you saw it
4. A fix they could make today

"It's not accessible" names none of them, so the designer cannot act on it.



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One design, explained two ways

Worked on contact preferences. In Activity 3 you do this for your own design.

For a tenant

You choose how we contact you about repairs: text, email or letter.

If you choose text, we ask for your mobile number on the same screen, and you can change it at any time.

What they can do and what happens next, in everyday words.

For a developer

Contact preference is a required single-choice radio group with no default; Text reveals a required mobile number field, validated on Save.

Errors render a summary linked to the field plus an inline message; the entered value is kept.

States, rules and behaviour, not colours.

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Activity 3: Review, improve and explain

Pairs, same breakout rooms

Review a partner's prototype, then improve yours.

1. Download your partner's .drawio from the breakout chat, open it and click through every link.
2. Check their screens against points 1, 2, 3, 6 and 7 of the standard (Activities document).
3. Write two review comments, each naming the screen, the point, what fails and a fix.
4. Make one change to your own design from the comments you received, and note it.
5. Write two sentences on your design for Joan and two for a developer; say them aloud.

Extension: see the Activities document

TIME
20 minutes working time

YOU NEED
Your partner's .drawio (breakout chat)
Activities document: the five-point check and tables

CORE OUTPUT
Two review comments, one change made, and your design explained twice.

Debrief: Read one comment aloud: could the designer fix it without asking you a question?

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Check your understanding

CHECK

1. A missing description is shown only by a red box border. Which point does it fail?

A Text contrast must be 4.5:1 or more against the background
B Colour must not be the only way the screen shows meaning
C Tap targets must be at least 24 by 24 CSS pixels
D Users must not re-enter information already given

2. On the portal's first screen, a tenant says they can smell gas. What should come next?

A The form continues, and the request is marked Emergency on submit
B The confirmation page shows a banner saying call us if it is urgent
C A photo becomes compulsory so that an adviser can judge the risk
D A stop screen: call 0800 111 999, or use Relay UK; nothing to submit

3. Which review comment could a designer act on without asking you a question?

A Screen 2 shows the error by a red border only: add the message in text
B The error screen is not very accessible yet, so please tidy it up
C I would choose a calmer colour scheme, because the orange feels too loud
D Nice work overall: the screens look clean, modern and very simple

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Skills covered today

KSB and official wording	In plain English	Where we covered it today
Part of S2: develop effective user interfaces	Designing and prototyping screens people can use on a phone, including errors and emergencies (building them comes in later modules)	Activities 2 and 3
Part of K6: how to contribute appropriately	Giving review comments a teammate can act on	Activity 3
Part of B4: with a positive attitude to inclusion & diversity	Designing for tenants with different needs and abilities	Activities 1 and 3
Part of B7: Communicates effectively ... to both a technical and non-technical audience.	Explaining your design to a tenant and to a developer	Activity 3

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Skills covered today (continued)

KSB and official wording	In plain English	Where we covered it today
Part of B8: creativity in their approach to solutions	Trying more than one layout before choosing	Activity 2

What to take away

1. Needs come from evidence: a persona line or a request_id, with a source.
2. One question per screen, visible labels, errors in words next to the field.
3. Emergencies come first, before sign-in: who to call, by voice or Relay UK.
4. A useful review comment names the screen, the point, what fails and a fix.

Before Session 3

Do the UI and UX Design, Wireframes and Prototypes - Consolidate Quiz (about 10 minutes).

Finish the Session 2 entry in your Reflective Log.

Read Use Cases, User Stories and Test Scenarios - Prepare.docx before Session 3.

Keep your .drawio file: Session 3 writes the rules your screens must follow.

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Reflective Log

Before you leave: about 5 minutes

Complete the Session 2 entry in Reflective Log.docx: one accessibility change you made to your wireframes, and why.

Then copy the key points into your OneFile timesheet reflection.

Keep it general: no personal or customer data, credentials or confidential material, and no employer data in AI tools your employer has not approved.

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Contact us

+44 (0)207 907 1120

info@firebrandtraining.co.uk

firebrand.training